

Field evaluation of laboratory techniques for predicting the ability of roots to penetrate strong soil and of the influence of roots on water sorptivity

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Abstract

The ability of two laboratory screening techniques to predict the abilities of roots of eight crop species to penetrate a compacted soil were evaluated and compared in a field experiment. A soil tilled to remove the effects of mechanical resistance was planted with the same species to serve as a control. Depth of root penetration, root density and the influence of the roots on the sorptivity of water were measured.

Roots of all species penetrated deeper in the deep tilled than compacted soils. There were differences in the ability of roots of the species to penetrate the compacted soil. Generally dicotyledonous species had more roots penetrating to depth in both the compact and deep tilled soils. Within the main species classifications, lupin and safflower (dicotyledons) and oats and barley (monocotyledons) had the highest penetration into the compacted soil.

Water sorptivities in the deep tilled soils were higher than those of the compact soil. Soil from planted treatments had higher sorptivities than soil which had not been planted. This is attributed to biopores left by the roots. Sorptivities of soils which had dicotyledonous species were generally higher than those of monocotyledons. The soil planted with safflower produced the highest sorptivity in the compacted layer (0.1–0.3 m).

A comparison of the accuracy of the two laboratory screening methods in predicting the field penetration of roots suggest that the method involving mechanical stress was better than that involving osmotic stress. Relative root diameter was found to be a better indicator of the penetration ability of roots than relative root elongation.

Introduction

There is extensive evidence to show that soil compaction limits root penetration and may restrict the ability of the plant to absorb water from the subsoil, particularly in semi-arid environments (Holloway, 1991; Voorhees, 1992). When roots cannot penetrate deeply because of compaction, crops are susceptible to drought and

consequent yield reduction occurs (Håkansson et al., 1987; Oussible et al., 1992). Deep tillage is one way to treat soil compaction to promote deep rooting (e.g. Coventry et al., 1987). However, crop responses to deep tillage have often been variable and the soil may recompact to densities equal to or even greater than before (Mead and Chan, 1988).

A potential alternative to deep tillage which

has not been adequately researched is the use of the 'biological plough'. The idea is to produce channels through the soil by using plants whose roots have superior ability to penetrate strong soil. When the roots that have penetrated subsequently decay, they leave behind biopores which can be used by the succeeding crop to enter the strong soil layer (Elkins, 1985). Roots preferentially grow in the biopores because there is much less mechanical impedance. Vertical biopores (root channels and earthworm tunnels) are therefore important to the development of roots at depth. The beneficial effects of pre-existing biopores in the subsoil on root development, water uptake and yield of plants have been clearly demonstrated (Elkins et al., 1977; Jakobsen and Dexter, 1988; Nambiar and Sands, 1992; Wang et al., 1986).

Although some authors have reported that roots of certain plants are more able to penetrate strong soil layers than others (Elkins et al., 1977; Henderson, 1989), there has been little systematic work done. Experience tells us that large genetic differences exist in root characteristics of crops suggesting that there is scope for screening large numbers of crop species and cultivars for the ability of their roots to penetrate strong soil. This requires the introduction of rapid techniques and criteria of selection which are not currently involved in breeding programs.

Two laboratory methods of screening species for the ability of their roots to penetrate strong soils have been developed (Materechera et al., 1991; 1992b). The two methods were used to select eight plant species for evaluation of their penetration into a compacted soil in the field. This paper reports the results of the field experiment and compares the accuracy of the two methods in predicting field penetration of roots of the species. The success of roots in penetrating and altering soil structure was further investigated by measuring the sorptivity of the soil to water. Another aim of the study was to test the hypothesis that differences in ability of roots of various species to penetrate strong soils in the field is a result of their rooting density under restricted conditions (Bennie and Burger, 1981).

Materials and methods

The laboratory methods

Method 1 (Mechanical stress)

This method has been described by Materechera et al., (1991). Briefly, it involved growing seedlings in compression chambers filled with fine siliceous sand at 0.20 kg kg^{-1} water content (-3.5 kPa potential) and either subjected to no mechanical stress or to a static uniaxial pressure of 14.6 kPa . The sand had a mean penetrometer resistance of 4.2 MPa , corresponding to an external mechanical stress on the roots of approximately 1.14 MPa . The plants were fed on nutrient solution and grown in a growth cabinet under well defined conditions for ten days.

Method 2 (Osmotic stress)

In the second method (Materechera et al., 1992b), seedling roots of the species were grown for 48 h in the same fine sand which was wetted with solutions of polyethylene glycol of $\text{MW} = 20,000$ with osmotic potentials of 0.0 , -0.25 , -0.5 and -1.0 MPa . The -1.0 MPa potential was used for comparison with mechanical stress.

In both methods, the diameters and elongation of the roots were measured at the end of the growth period. For comparison between species, the diameters and elongations of the stressed plants were expressed relative to those of unstressed plants grown under similar environments to obtain relative root diameter (RRD) and relative root elongation (RRE). Species were ranked according to these characteristics. Four species from the bottom rankings; barley (*Hordeum vulgare* cv Galleon), oats (*Avena sativa* cv Dolphin), ryegrass (*Lolium rigidum* cv Wimmera) and wheat (*Triticum aestivum* cv Kite) and four species from the top rankings; faba bean (*Vicia faba* cv Fiord), lupin (*Lupinus angustifolius* cv Gungurru), safflower (*Carthamus tinctorious* cv Gilla) and pea (*Pisum sativa* cv Early Dunn) were selected for evaluation in the field. The species at the top of the rankings had higher RRD and RRE while those at the bottom had lower values.

The field experiment

The field experiment was located at Roseworthy in South Australia (lat. 28°30'S, long. 115°7'E). The climate is mediterranean with cool, wet winters (June to August) and warm, dry summers (December to March). The average annual rainfall is 450 mm most of which falls during the winter and early spring months. The area supports mixed farming practices and is dominated by cereal/legume-based pasture rotations. The site had been under intensive cultivation with conventional tillage equipment for the past 25 years and a distinct layer of high strength (mean penetrometer resistance 3.0 MPa at 0.81 plastic limit; bulk density 1.6 Mg m⁻³) had developed below the tilled layer (between 0.1 and 0.3 m depth).

The surface soil is a well drained fine sandy loam and is a member of the red-brown earths. It is classified as Dr 2.43 (Northcote, 1979) or as a Calcic Rhodoxeralf (Soil Survey Staff, 1975). Soil tilled to a depth of 0.3 m (deep tilled) to reduce mechanical resistance (penetrometer resistance 0.2 to 0.5 MPa) was sown with the same species to serve as a control for soil strength. Deep tilling was done in May (1990) with a John Deere 4280 tractor pulling a John Shearer 14 tyne trash worker with narrow points. Some properties of the soils at different depths are presented in Table 1.

The experimental design was a split-plot with four replications in a randomised block layout. The main plots were the two compaction treatments (compacted and deep tilled soils) and subplots were four blocks (20 × 2 m) where the nine treatments (8 species and a non-planted control) were randomised in microplots (2 × 2 m). Seeds were broadcast on the flat seedbeds in June 1991 at a rate sufficient to produce 25 plants per m² for all species. Although this planting density was not optimum for the roots of all the species to explore the soil, it was ideal for both the root sampling scheme used and for comparing root penetration. All plots received one application (before sowing) of a phosphate fertiliser (19:30:0) which was broadcast onto the plots at a rate of 200 kg ha⁻¹.

Table 1. Some properties of the soil used

Property	Depth (m)		
	0.0–0.1	0.1–0.2	0.2–0.3
Particle size distribution (% w/w)			
2000–60	80.6	30.6	47.1
60–2	17.9	65.1	32.0
<2 µm	1.5	4.3	20.9
Water content (kg kg ⁻¹)			
Plastic Limit	0.19	0.16	0.20
Liquid Limit	0.29	0.27	0.36
pH (1:2.5 soil:water)	7.32	7.78	8.06
Organic carbon (%)	1.60	1.03	0.76
Bulk density (Mg m ⁻³)			
Compacted	1.16	1.57	1.43
Deep tilled	1.14	1.25	1.32
Penetrometer resistance (MPa)			
Compacted	0.26 ± 0.02	2.97 ± 0.26	2.36 ± 0.24
Deep tilled	0.04 ± 0.01	0.87 ± 0.03	1.04 ± 0.07

Root penetration measurements

Plants were harvested in November 1991 when all the species had matured. At that time, roots were sampled to determine the depth to which they had penetrated into the profile. Sampling tubes 10 cm in diameter were hydraulically driven into the soil to a depth of 2 m. The sampling tube was placed on top of a single plant. The soil core was carefully removed from the tube by pushing with a metal rod. The soil was cut into 10 cm segments. Each segment was then broken in the middle to reveal a cross section. The numbers of fresh roots, observable with the naked eye were counted on both faces. The simplicity of this 'core-break' method allowed rapid sampling and counting of roots (Drew and Saker, 1980). Two cores were extracted from each plot.

Root count (number) at each depth were used as a parameter of root density. The number of roots in the 0.0–0.1 m depth was considered to be 100 percent and the numbers found at other depths were expressed as proportions of these and multiplied by 100 to obtain the values (%) of root density reported in this paper. Thus, the

values of root density for other layers below 0–0.1 m indicate the percentage of roots which penetrated to or beyond that depth. Relative root density (RDR) was calculated to test the hypothesis that the differences in the ability of the various species to penetrate the compacted soil was related to their rooting density under unrestricted conditions of growth (i.e. deep tilled soil). This was done by expressing the root density in each depth of the compacted soil as proportion of those of the same species in the deep tilled soil i.e. $RDR = d_c/d_d$ where d_c and d_d are the root densities in the depths of the compacted and deep tilled soils respectively.

The accuracy of the two screening methods in predicting the field performance of roots was tested by examining the agreement in the rankings of the species with respect to RRE, RRD (from laboratory tests) and RDR (from the field test). This was done by using Spearman's rank correlation coefficients (Gibbons, 1974).

Sorptivity measurements

Macropores formed by the roots during penetration were examined by measuring water sorptivity of the soil with disc permeameters (White and Perroux, 1987). Sorptivity was used as a quick and easy method to measure the field hydraulic property of the soil to indicate treatment-induced changes in the ability of the soil to absorb water. Sorptivity is useful not only as a component of the infiltration equation, but also as an indicator of soil structure (White and Sully, 1987). It is in the latter sense that the measurements are used in this paper. Disc permeameters do not disturb the surface soil and the flow of water through macropores can be controlled. Measurements were made in summer (April 1992) when the soil was very dry. This was five months after plants were harvested and by this time, most of the roots had decayed to leave the biopores. Measurements were made on randomly selected positions in each plot under ponded conditions (0.05 kPa) and at a water supply pressure of -0.1 kPa, which is equivalent to a nominal pore size of 3 mm. Thus, only pores of nominal diameter 3 mm or less contributed to the flow of water (Perroux and White, 1988). Sorptivity was calculated from the early time data of cumulative infiltration measured at

depths of 0.0–0.1, 0.1–0.2, 0.2–0.3 and 0.3–0.4 m. Two measurements were made at each depth for each treatment in both the compacted and deep tilled soils.

Results

Depth of root penetration and root density of the plants in the compacted and deep tilled soils are presented in Figure 1. The densities of roots of all species in the 0.1–0.3 m depth of the deep tilled soil were higher (mean 86.9%) than those of the compacted soil (mean 45.9%). This means that there was a higher percentage of roots from the surface layer which penetrated to or beyond this depth in the deep tilled than compacted soils. The higher penetration of roots in the deep tilled soil is an obvious effect of reduced penetration resistance in this soil compared with the compacted soil.

In the compacted soil, roots of dicotyledonous species except faba bean generally penetrated deeper into the profile than monocotyledons. The maximum depths to which roots had penetrated this soil were 1.5 m for the dicotyledonous species and 0.90 m for the monocotyledons. A higher proportion of roots from the top layer in dicotyledons penetrated through the compact layer (0.1–0.30 m) than in monocotyledons. The average root densities were 28.3% and 63.4% for monocotyledons and dicotyledons respectively. Lupin and safflower had the highest root densities to depth among dicotyledonous species while oats and wheat were highest among monocotyledons.

The RDRs of the species are presented for each depth in Table 2. The RDR for most species decreased in compact layer (0.1–0.3 m), and increased again below about 0.5 m for most species. The lower RDR values in the top 0.5 m show that there was a higher proportion of roots penetrating this depth in the deep tilled than compacted soil. This could be a direct effect of the reduced resistance experienced by the roots in the deep tilled soil. The results indicate that a higher proportion of roots of dicotyledonous species than monocotyledons penetrated the compacted soil and that there were differences between the species within the groups. Lupin

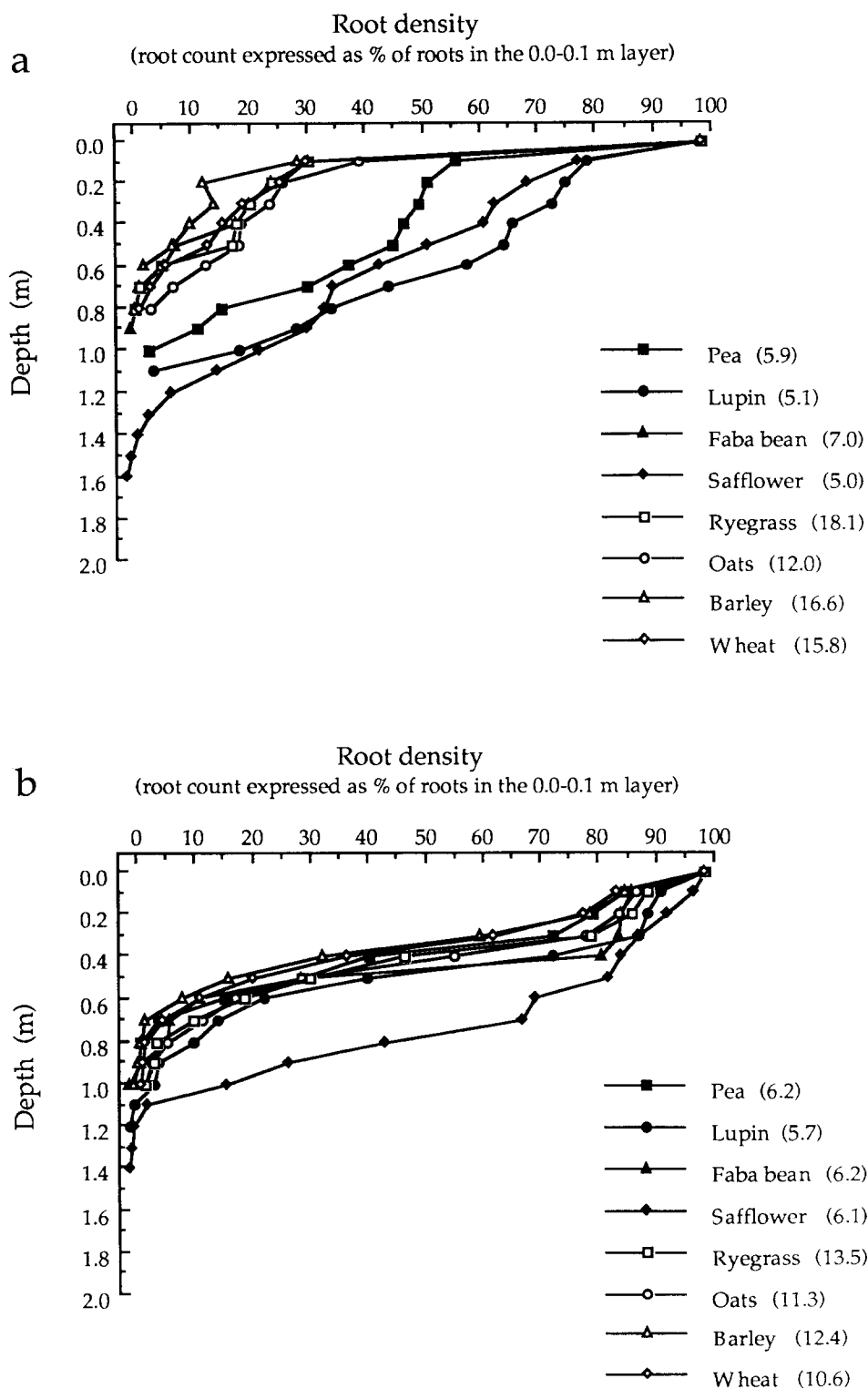


Fig. 1. Root density (root counts in each layer expressed as % of root in the 0.0–0.1 m layer) in the profiles of the compacted **a** and deep tilled **b** soils at maturity. Numbers in parenthesis are the mean ($n = 16$) counts of roots in the surface (0.0–0.1 m) layer for each species.

Table 2. Relative root densities (RDR)^a of the species at different depths

Depth (m)	RDR							
	Faba bean	Lupin	Pea	Safflower	Barley	Oats	Ryegrass	Wheat
0.0–0.1	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
0.1–0.2	0.36	0.87	0.66	0.80	0.35	0.46	0.35	0.37
0.2–0.3	0.30	0.85	0.65	0.75	0.17	0.29	0.29	0.34
0.3–0.4	0.25	0.84	0.69	0.72	0.26	0.34	0.27	0.33
0.4–0.5	0.24	0.91	1.16	0.73	0.34	0.35	0.40	0.45
0.5–0.6	0.29	1.59	1.54	0.63	0.48	0.66	0.59	0.67
0.6–0.7	0.54	2.54	2.34	0.62	0.38	0.76	0.32	0.57
0.7–0.8	0.37	2.92	5.75	0.53	np	0.63	0.24	0.80
0.8–0.9	0.26	3.15	7.04	0.78	np	0.64	0.40	0.78
0.9–1.0	0.57	5.71	np	1.13	np	np	np	np
1.0–1.1	np ^b	4.30	np	1.35	np	np	np	np
1.1–1.2	np	3.77	np	4.71	np	np	np	np
1.2–1.3	np	np	np	6.67	np	np	np	np
1.3–1.4	np	np	np	5.71	np	np	np	np
1.4–1.5	np	np	np	10.5	np	np	np	np
1.5–1.6	np	np	np	np	np	np	np	np
1.6–1.7	np	np	np	np	np	np	np	np

^a Root density in compacted soil as proportion of that in deep tilled soil.

^b np = roots of the species did not penetrate to this depth.

and safflower had the highest RDRs among the dicotyledons, while oats and wheat were top among the monocotyledons.

A summary of the rankings of the species for RRE, RRD (from laboratory tests) and RDR (from field) is presented in Table 3. The results show that the dicotyledonous species (except faba bean) had consistently higher rankings in all the tests than monocotyledonous species. There is however interchanging of the rank positions for the species in the different tests. This suggests that the characteristics RRE and RDR are only a rough guide to the actual ability of the roots to penetrate strong soil. Correlation co-

efficients of rankings of the species for RRE, RRD and RDR were 0.53*, 0.74**, 0.51* and 0.67** for RRD_o v RDR, RRD_m v RDR, RRE_o v RDR and RRE_m v RDR respectively. The subscripts o and m stand for osmotic and mechanical stresses respectively and significance levels are given as * ($p \leq 0.05$) and ** ($p \leq 0.01$).

The sorptivities of water for the soils are presented in Figure 2. The values were generally higher in the deep tilled than compacted soils, presumably due to the deep tilling operation making the soil more porous. Soils with plants had higher sorptivities than those without plants presumably due to biopores left by the roots of

Table 3. A comparative summary of the rankings of the species based on RRE, RRD (from the laboratory tests) and RDR (from the field test)

Rank position	Laboratory				Field
	Mechanical stress		Osmotic stress		Root penetration
	RRE	RRD	RRE	RRD	RDR ^a
1	Lupin	Pea	Faba bean	Faba bean	Lupin
2	Pea	Lupin	Pea	Lupin	Safflower
3	Faba bean	Safflower	Lupin	Pea	Pea
4	Safflower	Faba bean	Safflower	Safflower	Oats
5	Ryegrass	Ryegrass	Oats	Ryegrass	Wheat
6	Wheat	Oats	Ryegrass	Oats	Faba bean
7	Oats	Wheat	Wheat	Wheat	Ryegrass
8	Barley	Barley	Barley	Barley	Barley

^a Ranking based on mean RDR of the species in the compact (0.1–0.3 m depth) layer.

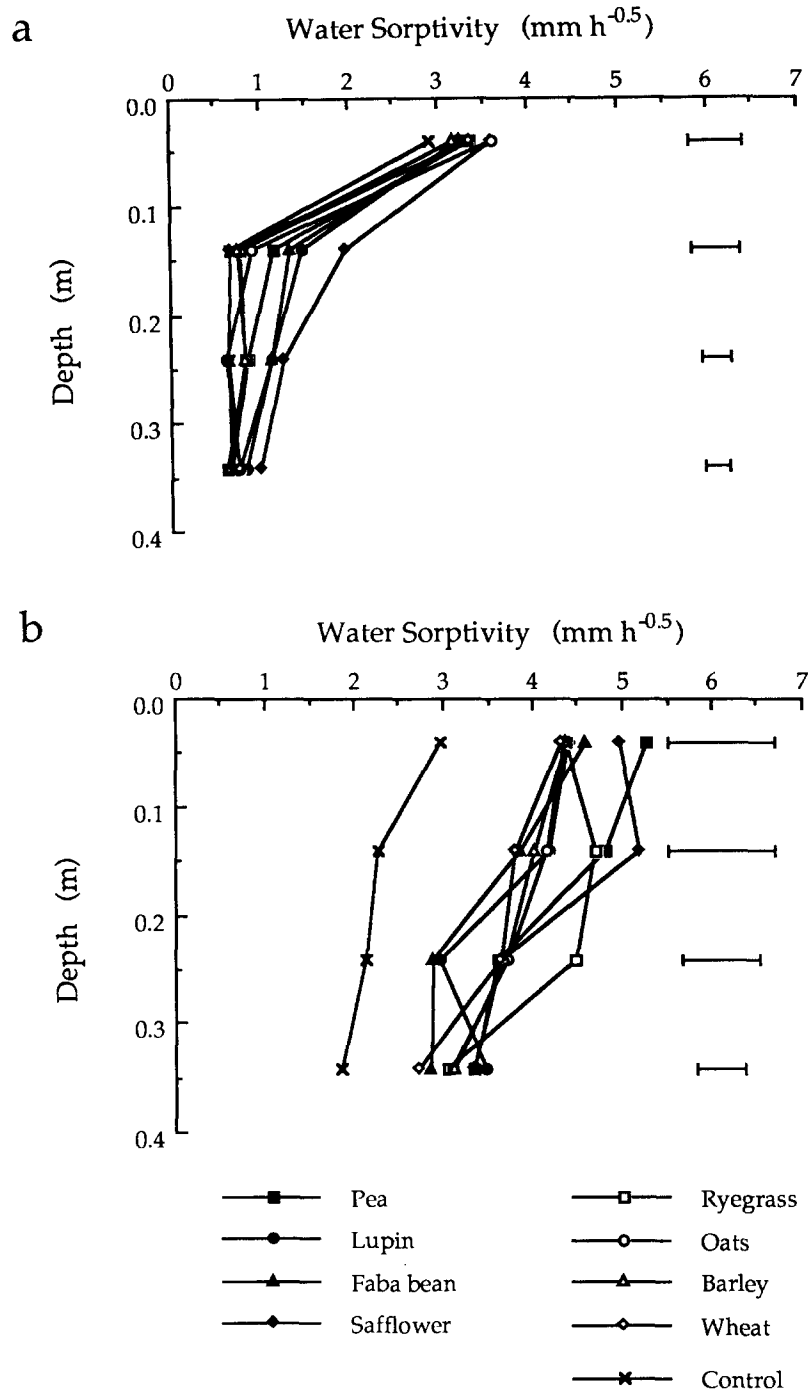


Fig. 2. Influence of plant species on water sorptivity of soils in the compacted **a** and deep tilled **b** soils at different depths. Bars are LSD ($p \leq 0.05$) for each depth.

the plants. The sorptivities of water through the compacted soil layers were generally higher in plots which had been planted to dicotyledonous

species than those with monocotyledons. Soils which were planted with safflower and oats had the highest sorptivities in the compact layer

Table 4. Water content (kg kg^{-1}) of the soil from the different treatments at the time of sorptivity measurements

Depth (m)	Compacted soil		Deep tilled soil	
	Planted	Non planted	Planted	Non planted
0.0–0.1	0.068 $\pm$.003	0.071 $\pm$.006 (0.070)a	0.066 $\pm$.002	0.065 $\pm$.004 (0.066)a
0.1–0.2	0.076 $\pm$.006	0.083 $\pm$.004 (0.080)a	0.069 $\pm$.005	0.072 $\pm$.006 (0.071)a
0.2–0.3	0.093 $\pm$.005	0.097 $\pm$.006 (0.095)b	0.085 $\pm$.006	0.089 $\pm$.007 (0.087)b
0.3–0.4	0.124 $\pm$.002	0.115 $\pm$.003 (0.120)c	0.101 $\pm$.003	0.116 $\pm$.002 (0.109)c
Mean	0.090	0.092	0.080	0.086

Values are means ($n = 36$) $\pm$ standard error for the four replicates. Numbers in parenthesis are the means for each depth. Means within a column followed by the same letter are not significantly different ($p \leq 0.05$) by the Tukey's test.

among dicotyledonous and monocotyledonous species respectively. In the deep tilled soil however, the sorptivities did not differ much with species. The variation in water content of the soils at the time of measuring sorptivity is presented in Table 4. Analysis of variance showed that soil water content did not differ significantly between compacted and deep tilled soils and also between planted and non planted treatments. There were however significant ($p \leq 0.05$) differences in water content of the soils at different depths.

Discussion

In terms of the absolute numbers of roots in the 0.0–0.1 m layer (Fig. 1), lupin, safflower and pea had more roots in the deep tilled soil than in the compacted soil whereas the other species had more roots in the compacted soil. Species which did not penetrate more into the compact layer had more roots in the surface layer. The implication is that roots of those species that were less well able to penetrate the compact layer were near the surface. Lupin, pea and safflower also had a higher proportion of roots in the compacted soil below 0.3 m than in the same depth of the deep tilled soil (hence higher RDRs). The implication might be that having successfully penetrated through the compact layer, the roots of these species were then effective at exploiting subsoil resources in the compacted soil. In the deep tilled soil, the plants did not so exploit the resources, presumably because they had no need. This could be a case of compensatory growth when plants with part of their roots growing in unfavourable environments may

produce more roots in the more favourable part of the soil than others where conditions are favourable throughout. The other five species (barley, faba bean, oats, ryegrass and wheat) did less well at penetrating the compact layer, and had a smaller proportion of roots below 0.3 m in the compacted soil than in the deep tilled soil.

As noted above, the number of roots which the plant had in the deep tilled soil did not seem to have influenced the penetration of the roots. Species which had large numbers of roots in the top (surface) layer did not have a higher penetration through the compact layer. For example faba bean (dicotyledon) and ryegrass (monocotyledon) had the highest numbers of roots in the surface layer of the deep tilled soil but did not have the highest proportions of roots penetrating through the compact layer. On the contrary, they had the lowest RDRs. This observation is in contrast with the conclusion of Bennie and Burger (1981) that the ability of the roots to penetrate strong soil is related to the numbers of roots a plant has in unrestricted conditions. In light of the findings from this study, it is suggested that the ability of roots to penetrate strong soil would be influenced by root numbers only if the soil in which they are growing has an abundance of cracks and pre-existing biopores. In this case, the species with large numbers of roots (mostly monocotyledons) would have a statistical advantage over those with fewer roots (dicotyledons) with regard to entry into these paths of low resistance (Dexter, 1986).

The compacted soil in this study was non-swelling and did not crack on drying. Moreover it did not have many visible pre-existing biopores. Thus, it is likely, that penetration of roots would have been influenced more by mecha-

nisms other than entry of roots into existing channels. Our earlier findings (Materechera et al., 1992a) suggest that root diameters could have significant influence on the ability of roots to penetrate strong soil. This has been related to the ability of the roots with bigger diameters to deform the soil more effectively than those with smaller diameters.

The possible effects of the root sampling procedure on RDRs need to be mentioned here. Taking samples directly underneath the plant may have overestimated root development for tap-rooted dicotyledons and underestimated that of monocotyledons particularly in the 0.0–0.1 m layer. The plant root densities directly underneath the plants of monocotyledonous species can be low due to the orientation of nodal roots. However, as counting was made on roots which were vertically oriented to the breakage face of the core, the error caused by the sampling procedure would have been greatly reduced.

In the compacted soil, growth of dicotyledonous species (with bigger roots) resulted in greater sorptivities than was the case for monocotyledons which in turn produced greater sorptivities than in the soil in which no plants had been grown. The greatest sorptivities in the compact soil were from plots in which safflower and lupins had been grown. The plots with pea were little different from the monocotyledons plots. Although sorptivity is influenced much by the antecedent water content of the soil, this is not considered to be the main reason for the differences observed between the plots with plants and those without plants. Table 4 shows that the water content of the soil in the two treatments did not differ significantly at the time when sorptivity was measured. The higher sorptivities of the soils with dicotyledons compared with those planted with monocotyledons is presumably a result of the bigger biopores created by the roots of the dicotyledonous species. Although diameters of the resulting biopores were not measured, it is logical to expect that the biopores left by safflower roots for example (mean diameter $1.12 \pm .14$ mm) would be bigger than those left by ryegrass (mean diameter $0.22 \pm .07$ mm).

It can be estimated from the proportions of roots penetrating the compact layer and the

planting density that the roots of safflower would leave about 100 biopores per m^2 while ryegrass would leave about 160 per m^2 . However, despite the fewer biopores safflower produced higher infiltration of water through the compact layer. This is consistent with the above explanation that the higher sorptivities associated with the safflower could be due to the size of the biopores created by its roots. As dicotyledonous species are normally grown at lower densities than the fine-seeded grasses, it seems that there will be few such pores created per unit area of ground. It may be worthwhile planting these plants at densities of several hundred per m^2 to increase the number of pores for root penetration. It is unlikely, especially under dryland conditions, that deep rooted species which do not have a commercial return in their own right could be justified economically and grown solely for their structural amelioration benefits. Future comparisons will need to consider the use of 'economic optimum' planting densities for each species.

A comparison of RRE and RRD as predictors of field performance of roots showed that the laboratory screening method involving mechanical stress had higher correlations with field penetration for both RRE and RRD than that involving osmotic stress. The better agreement in the rankings of the field result with mechanical stress method could be due to similarity in conditions under which the roots grew in the two situations. It could also be related to the efficiency of the plant roots to osmoregulate in the two stress conditions. Greacen and Oh (1972) found that pea roots osmoregulated 100% against total soil water potential down to -1.5 MPa potential while it osmoregulated by 70% efficiency against mechanical resistance. The results from this work however suggest that the method involving mechanical stress is more suitable for screening species for field performance of their roots. In both methods however, RRD had higher correlations with the field penetration than RRE. This would suggest that RRD is a better indicator of the ability of roots to penetrate strong soil than RRE.

It would seem that screening and probably breeding of plant species and cultivars for the ability of their roots to penetrate strong soil layers could become a valuable management tool

with which to improve the drainage of water in soils with compacted subsoil. This would also enable plants to gain access to water and nutrients in the subsoils. Species with superior ability to penetrate strong soil can then be incorporated into existing cropping systems. Techniques developed in this study could help to identify and select species with this potential.

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